DATA VISUALIZATION (CSE355) CASE STUDY

Phase 1 — Primary Review Report

Global Marine Microplastic Pollution

Field	Detail
Case Study Title	Global Marine Microplastic Pollution: An Ethical, Bias-Aware Visualization Framework for Hotspot Detection
Domain	Environment / Marine Pollution
Dataset	NOAA NCEI Global Marine Microplastics Database (subset, 1989–2016)
Name and roll No	K Ganesh Giridhar (CB.SC.U4CSE24519)

1. Dataset Source & Description

This study uses the **NOAA National Centers for Environmental Information (NCEI) Global Marine Microplastics Database**, a curated, open-access repository that aggregates marine microplastic observations contributed by research institutions and citizen-science programs worldwide. The full collection spans 1972 to the present; the working subset for this case study contains roughly **29,000+ observations across 32 attributes, covering 1987–2022** and every major ocean basin.

A defining and analytically critical property of this database, confirmed from the official NCEI metadata, is that **microplastic concentrations are reported in different units depending on the marine setting** — pieces/m^3 (ocean and coastal water), pieces per 10 minutes (surface visual / nurdle counts), and pieces kg^{-1} dry weight (sediment and beach). Each marine setting follows its own normalized sampling protocol, so the raw measurement column is ***not directly comparable across settings***. This single fact drives the ethical-visualization strategy of the entire project.

Provenance: The subset blends professional oceanographic expeditions (e.g., Sea Education Association surface tows analysed by Law et al. 2010 in the North Atlantic and Law et al. 2014 in the eastern Pacific) with global citizen-science grab sampling (Barrows et al. 2018 via Adventure Scientists) and additional contributors. Each record carries a short reference, organization, and NCEI accession number, enabling source-level reliability weighting.

2. Problem Statement

Every minute, roughly one garbage-truck load of plastic enters the ocean, and more than 11 million metric tonnes flow in each year — a figure projected to triple by 2040 without intervention. Through sunlight, wave action, and microbial breakdown, this debris fragments into **microplastics (particles < 5 mm)** that now contaminate every ocean basin, from subtropical-gyre surfaces to

Arctic seafloor sediments. These particles infiltrate marine food webs, adsorb persistent organic pollutants, and have been detected in commercial fish, shellfish, and even human blood — making microplastic contamination both an ecological crisis and a public-health and coastal-economic threat.

The core analytical problem is not a shortage of data but its **fragmentation and methodological incompatibility**. Observations are collected through fundamentally different protocols (neuston/manta net tows, subsurface grab samples, sediment cores, timed beach counts), reported in three incompatible units, and distributed across highly uneven geographies and decades. As a result, decision-makers risk **mistaking a heavily-sampled region for a genuine pollution hotspot**, and misreading rising observation counts as rising pollution — misallocating scarce cleanup and regulatory resources.

This case study transforms 29,000+ heterogeneous observations into an **ethical, interactive visualization framework** that separates really spatial–temporal pollution signals from sampling artefacts, and demonstrates explicitly how sampling effort, unit mixing, and chart-design choices can distort public understanding of environmental risk if left unaddressed.

3. Objectives of the Analysis

The analysis pursues ten specific, measurable objectives. Each state what will be produced, the concrete success criterion, and the analytical question it answers — making progress verifiable at later review stages.

	Objectives	Measurable success criterion	Core question answered
1	Map global spatial distribution	Plot 100% of geo-valid records on a global map with ocean-basin encoding.	Where has pollution been observed, and where are the critical data gaps?
2	Normalize for sampling effort	Produce an effort-adjusted hotspot view (concentration ÷ sample density per grid cell).	Which apparent hotspots are real vs. artefacts of oversampling?
3	Decompose temporal trends	Chart annual mean concentration against annual sample count, 1987–2022, split by method.	Is pollution genuinely worsening, or are we simply measuring more?
4	Compare ocean basins	Quantify median, IQR, and outliers of concentration for each basin (unit-matched).	Which ocean basins carry the highest pollution loads?
5	Compare marine settings	Contrast concentration-class distributions across water, sediment, and beach settings.	Which environmental compartment is most contaminated relative to its baseline?
6	Assess sampling-method & mesh bias	Measure the mean-concentration gap between methods within each ocean and vs. mesh size.	How much observed variation is a detection artefact rather than real pollution?
7	Audit unit heterogeneity	Verify that no visualization mixes units on one numeric axis; establish class-based comparison.	How can incompatible units be compared without misleading the viewer?

8	Profile concentration-class severity	Report the proportion of Very-Low→Very-High observations by ocean and era.	What share of observations indicates genuinely concerning pollution levels?
9	Analyse provenance & effort distribution	Break down records by organization and professional vs. citizen-science source.	How reliable and how geographically complete is each data source?
10	Deliver an interactive Power BI dashboard	Ship a filterable dashboard (ocean, setting, method, year, class) with KPI cards and drill-down.	How can these insights be communicated to drive policy action?

4. Description of Important Attributes

The subset holds ~29,000 records across 32 attributes. The table below profiles the analytically load-bearing fields their data type, description, and role in the plan.

Attribute	Type	Description	Analytical role
Sample Date	DateTime	Collection date, 1987–2022 (text form e.g. 10/18/1991 5:00:00 AM).	Temporal trend axis (O3)
Latitude / Longitude	Numeric	Decimal degrees: lat ≈ −54° to +81°, global longitude coverage.	Spatial mapping (O1, O2)
x, y	Numeric	Pre-computed Web Mercator projected coordinates.	GIS-ready rendering
Ocean	Categorical	Atlantic, Pacific, Indian, Arctic, Southern.	Primary spatial grouping (O4)
Region / Subregion	Categorical	Named sub-basins (Caribbean, Mediterranean, Gulf of California, Tasman…).	Granular stratification
Marine Setting	Categorical	Ocean water, Ocean sediment, Beach, Nurdle Patrol.	Compartment segmentation (O5)
Sampling Method	Categorical	Neuston/Manta/AVANI net, Grab sample, Hand picking, Megacorer, ROV.	Method-bias variable (O6)
Mesh Size (mm)	Numeric	Aperture from ~0.00045 to 1.0 mm; smaller mesh captures more particles.	Detection-sensitivity control (O6)
Microplastics Measurement	Numeric	Core concentration value (heavily right-skewed).	Primary quantitative target
Unit	Categorical	pieces/m³, pieces/10 mins, pieces kg⁻¹ d.w. — varies by setting.	Unit-aware comparison gate (O7)
Concentration Class Range	Ordinal	Binned numeric intervals (unit-specific).	Ordinal classification
Concentration Class	Ordinal	Very Low, Low, Medium, High, Very High.	Cross-unit comparison & color coding (O8)
Organization	Categorical	SEA, 5 Gyres, Adventure Scientists, Marine Scotland Science, etc.	Provenance & effort (O9)

Short Reference / DOI	Text	Source citation (Law 2010/2014, Barrows 2018…).	Reliability weighting
Depth fields	Numeric	Ocean-bottom, water-sample, sediment depth; often null for surface tows.	Environmental context
Country / State / Beach	Categorical	Political geography; largely empty for open-ocean records.	Sparse contextual metadata

5. Initial Data Quality Assessment

A structured review identifies eight quality dimensions, each tagged with a severity and a concrete mitigation that shapes every downstream visualization decision.

Quality dimension	Finding	Severity	Mitigation strategy
Unit incompatibility	Three coexisting units (pieces/m³, /10 min, kg⁻¹ d.w.) tied to marine setting — direct numeric comparison across settings is invalid.	Critical	Never mix units on one axis; use Concentration Class for cross-setting comparison.
Sampling-effort bias	A few basins/gyres (esp. North Atlantic) dominate record counts; Indian & Southern Oceans are sparse.	High	Always show effort alongside concentration; annotate data deserts.
Method heterogeneity	Neuston net (coarse mesh, surface) vs. grab/core (fine mesh, subsurface/sediment) capture different particle populations.	Critical	Stratify every analysis by method/mesh; never aggregate silently.
Structural missingness (MNAR)	Region, Country, State, Beach, Volunteers, Collecting Time empty for open-ocean records — not random.	High	Treat as structural absence; use only where populated; no blanket imputation.
Temporal sparsity	Pre-2000 limited to a few SEA voyages; dense coverage only post-2008; grab samples appear ~2013.	Medium	Show trends with gaps annotated; separate method eras.
Right-skewed distribution	Most records are Very Low/Low; High/Very High are rare and clustered.	Medium	Log-transform for statistical plots; ordinal classes for communication.
Duplicate coordinates	Repeat tows at fixed stations share identical lat/long, risking overplotting & overweighting.	Low	Aggregate/flag duplicates before spatial aggregation.
Text date format	Dates stored as text with a time component.	Low	Parse to datetime; derive year/month for aggregation.

Overall verdict: with ~29,000 records spanning three decades and all major basins, the dataset is highly relevant and more than sufficient in scale for global inferential visualization. The two critical risks — unit incompatibility and method heterogeneity — are addressable by design rather than by discarding data, which is precisely what elevates this project into an ethical-visualization exercise.

6. Proposed Data Cleaning Approach

Cleaning is sequenced so that each step de-risks a specific quality finding above:

- **Parse & standardize dates:** convert text timestamps to datetime; derive Year, Month, and a pre-2008 / post-2008 era flag for trend eras.
- **Validate coordinates:** drop or flag records with lat/long outside valid ranges or with (0,0) placeholders; reconcile against the pre-projected x/y.
- **Segregate by unit:** partition the measurement column by Unit so numeric aggregation only ever occurs within a single unit; all cross-unit views fall back to Concentration Class.
- **Handle missingness as MNAR:** keep sparse geographic/volunteer fields as explicit "not recorded" rather than imputing; restrict country-level views to populated subsets.
- **De-duplicate replicates:** define a rule for identical coordinate–date–reference tuples (aggregate to mean for mapping; retain individually for method comparison).
- **Standardize categoricals:** normalize inconsistent labels/casing in Ocean, Marine Setting, Sampling Method, and Organization; order Concentration Class as an ordered factor.
- **Transform for skew:** create a log1p(measurement) field for statistical distribution plots while preserving raw values for reporting.
- **Engineer effort features:** compute records-per-grid-cell and records-per-region to power the effort-normalized hotspot view (O2).

7. Selected Visualization Tools

Three complementary tools are used, exceeding the two-tool minimum. Each is chosen for the job it does best rather than for coverage alone.

Tool	Why it is used	What it produces here
Matplotlib	Fine-grained control for publication-quality static figures and annotations.	Dual-axis temporal trends, effort-bias maps, stacked-proportion bars, and the annotated misleading-vs-ethical figure.
Seaborn	High-level statistical grammar for distribution and matrix views.	Boxplots, violin plots, regression scatter, and mean-concentration heatmaps that expose method/ocean variance.
Power BI	Interactive, filterable delivery for non-technical stakeholders.	Integrated dashboard: global map, KPI cards, slicers (ocean/setting/method/year/class), and drill-down trend panels.

8. Proposed Visualization Plan

Ten visualizations cover the full analytical spectrum — spatial, temporal, categorical, statistical, methodological, and ethical — and collectively use all three tools plus every core chart type (line, scatter, histogram, boxplot, bar, heatmap, violin, map). The summary table is followed by detailed cards for each.

	Title	Chart type	Tool	Expected insight
V1	Global Microplastic Observation Map	Geographic bubble map	Power BI	Reveals gyre accumulation and exposes Indian/Southern Ocean data deserts.
V2	Effort-Normalized Hotspot Map	Grid bubble (size=count, color=mean)	Matplotlib	Separates true hotspots from oversampled transects.
V3	Concentration vs. Sampling Effort Over Time	Dual-axis line + bar	Matplotlib	Decouples pollution trend from monitoring-effort growth.
V4	Concentration Distribution by Ocean Basin	Histogram + boxplot (log)	Seaborn	Whether the Atlantic median is real or a sampling artefact.
V5	Concentration by Marine Setting	Violin plot	Seaborn	Which compartment is most contaminated vs. its baseline.
V6	Sampling Method × Ocean Mean Heatmap	Annotated heatmap	Seaborn	Quantifies method bias per ocean, unit-matched.
V7	Mesh Size vs. Reported Concentration	Regression scatter	Seaborn	Confirms finer mesh inflates counts — a detection confound.
V8	Concentration-Class Composition by Ocean & Era	Stacked proportion bar	Matplotlib	Severity mix, normalized for unequal sample sizes.
V9	Misleading Chart + Ethical Redesign	Truncated-axis bar vs. corrected	Matplotlib	Demonstrates how axis tricks distort perceived risk.
V10	Integrated Interactive Dashboard	Map + KPI + slicers + trends	Power BI	Turns a uniform 'red ocean' into an interrogable decision tool.

9. Expected Business & Analytical Insights

Insight 1 — Sampling bias dominates the spatial narrative. Repeated North-Atlantic voyages make that basin look most polluted when it is mainly most sampled; effort normalization (V2) rebalances the picture.

Insight 2 — Temporal growth reflects research expansion. Record counts grow sharply post-2010 and finer-mesh grab sampling arrives ~2013 — separating count from mean concentration (V3) prevents reading 'more measurement' as 'more pollution'.

Insight 3 — Subtropical gyres are genuine hotspots. After effort correction, persistent High/Very-High zones align with the North Atlantic and North Pacific gyres (~20°–40° N), validating accumulation theory and flagging cleanup priority areas.

Insight 4 — Method & mesh drive apparent differences. Finer mesh detects more (esp. microfibers) while coarse neuston nets show higher peaks but more zeros — a detection-limit paradox (V6, V7) that must not be misread as geography.

Insight 5 — Units make cross-setting comparison impossible without classes. Water, sediment, and timed-count data cannot share a numeric axis; the ordinal Concentration Class is the only ethical basis for cross-setting comparison (V5, V8).

Insight 6 — Citizen science adds breadth but needs transparency. Volunteer grab samples extend coverage into the Indian, Arctic, and Southern Oceans, but their distinct protocol requires clear labelling to avoid misleading comparisons (V1, V10).

Conclusion

This Phase 1 report establishes the analytical foundation for subsequent implementation. The dataset's richness — three decades, all major basins, multiple marine settings, and both professional and citizen-science methodologies — provides an ideal platform for demonstrating how responsible, bias-aware visualization transforms fragmented environmental data into evidence for action.